EARLY PREDICTION OF POOR AIR QUALITY IN INDIAN CITIES USING MACHINE LEARNING

A Comparative Analysis of Pollutant and Temporal Factors

Course End Project report submitted in partial fulfillment of the Requirements for the Award of the Degree of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

By

Aditya Kollapudi – 160124771302

Akhil Kandibanda – 160124771042

Under the Mentorship of

Mrs. V. Vasuki Rohini Devi



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY

HYDERABAD – 500075

October 2026

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

APPLIED PREDICTIVE ANALYTICS (22ADE57N)

CERTIFICATE

This is to certify that the course end project report entitled **"Early Prediction of Poor Air Quality in Indian Cities Using Machine Learning"** being submitted by

Name	**Roll No.**
Aditya Kollapudi	160124771302
Akhil Kandibanda	160124771042

in partial fulfillment for the award of the Degree of Bachelor of Engineering in Artificial Intelligence and Data Science to the CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD, carried out under my mentorship and supervision during the year 2026–27. The results presented in this report have not been submitted to any other university or institute for the award of any degree.

Mrs. V. Vasuki Rohini Devi
Assistant Professor

ABSTRACT

Air pollution is one of the most serious public-health problems in urban India. This project, titled "Early Prediction of Poor Air Quality in Indian Cities Using Machine Learning", analyses the public Kaggle dataset *Air Quality Data in India (2015–2024)* (ankushpanday1/air-quality-data-in-india-2015-2024). The file used, city_hour.csv, contains 438,245 hourly records for five cities (Delhi, Mumbai, Chennai, Kolkata and Bangalore) with twelve pollutant measurements, a numerical AQI and an AQI category.

The study follows a complete predictive-analytics workflow: automatic column discovery, data-quality checks, outlier analysis, calendar and season features, twelve exploratory visualizations and leakage-safe scikit-learn pipelines. Linear Regression and a Decision Tree were trained to predict AQI on 350,596 records and tested on 87,649 records.

The results are unusual but clear. Every pollutant has a correlation of 0.00 with AQI, all cities, months, seasons and years average close to 250, and the AQI categories are almost perfectly balanced. Linear Regression reached $R^2 = -0.0003$ (no better than predicting the mean) and the Decision Tree $R^2 = -1.04$. The numbers match what statistics predicts for independent, uniformly distributed random values, which indicates that this dataset is synthetic and does not follow the CPCB AQI formula. The report presents the evidence, explains why the models fail, and states what this means for the remaining experiments.

Keywords: Air Quality Index, PM2.5, exploratory data analysis, regression, data validation, synthetic data, predictive analytics.

REPORT ORGANISATION

Section	Contents
1	Introduction, problem statement, objectives and research questions
2	Background: the Indian National AQI
3	Dataset source, structure and data dictionary
4	Data preparation and data-quality results
5	Exploratory data analysis and visualizations
6	Predictive modelling and evaluation
7	Is the dataset realistic? Validation against the CPCB formula
8	Answers to the research questions
9	Limitations and future scope
10	Conclusion
11	Tools, reproducibility and references

1. INTRODUCTION

1.1 Background

Air pollution is the presence of harmful substances in the air at levels that can damage health and the environment. The main urban pollutants are particulate matter (PM2.5 and PM10), gases such as nitrogen dioxide (NO_2), sulphur dioxide (SO_2), carbon monoxide (CO) and ozone (O_3), and volatile organic compounds such as benzene, toluene and xylene. Traffic, industry, construction dust and seasonal crop-residue burning make air quality a major public-health concern in Indian cities.

The **Air Quality Index (AQI)** turns many pollutant readings into one number and a category (Good, Satisfactory, Moderate, Poor, Very Poor, Severe), which makes health risk easy to communicate. Machine learning can learn how pollutants, location and time relate to AQI and can support an early-warning system for poor air quality.

1.2 Problem Statement

Air pollution varies across Indian cities and over time. Predictive analytics can identify poor air-quality conditions and show which factors carry the most predictive information. The central question of this project is: **can machine learning estimate AQI and give early warning of poor air quality from pollutant, city and time information?** All relationships reported are predictive associations, not causal effects.

1.3 Project Objectives

- Predict numerical AQI and AQI categories from pollutant and temporal information.
- Identify the most important variables for predicting AQI.
- Analyse pollution patterns across cities, months, seasons and years.
- Test whether multiple pollutants and city information improve prediction.
- Build an early-warning classifier for Poor, Very Poor and Severe conditions.
- Test whether historical observations can predict next-period AQI.

1.4 Research Questions

RQ	Question
RQ1	Can machine learning predict AQI from pollutant concentrations and other features?
RQ2	Can machine learning predict AQI categories?
RQ3	Which pollutants are the strongest predictors of AQI?
RQ4	Do multiple pollutants improve prediction compared with PM2.5 alone?
RQ5	Does adding City information improve AQI prediction?
RQ6	Which Indian cities have the highest average pollution levels?
RQ7	How does air quality vary across months and seasons?
RQ8	How has air quality changed over the years in the dataset?
RQ9	Can poor or severe air-quality observations be identified with machine learning?
RQ10	Can previous observations predict future (next-period) AQI?

2. BACKGROUND: THE INDIAN NATIONAL AQI

India's National AQI, defined by the Central Pollution Control Board (CPCB), is calculated from pollutant concentrations. For each pollutant a sub-index I_p is obtained by linear interpolation between concentration breakpoints, and the overall AQI is the largest sub-index:

$$I_p = \frac{I_{hi} - I_{lo}}{B_{hi} - B_{lo}} (C_p - B_{lo}) + I_{lo}, \qquad \text{AQI} = \max_p I_p \tag{1}$$

where C_p is the concentration, B_{lo}, B_{hi} are the breakpoints around it and I_{lo}, I_{hi} the matching index values. In real data, AQI therefore rises with the dominant pollutant (usually PM2.5 or PM10 in India), and the category follows directly from the AQI value.

AQI range	Category	Possible health impact
0 – 50	Good	Minimal impact
51 – 100	Satisfactory	Minor breathing discomfort to sensitive people
101 – 200	Moderate	Discomfort to people with lung or heart disease, children and older adults
201 – 300	Poor	Breathing discomfort to most people on prolonged exposure
301 – 400	Very Poor	Respiratory illness on prolonged exposure
401 – 500	Severe	Affects healthy people; serious impact on people with existing disease

Table 1. CPCB AQI categories.

3. DATASET DESCRIPTION

3.1 Dataset Source

The dataset was downloaded from Kaggle with kagglehub. The download contains five CSV files: station_day.csv (3.93 MB), stations.csv, station_hour.csv (0.84 MB), city_hour.csv (46.25 MB) and city_day.csv (1.76 MB). The notebook selects the largest file, city_hour.csv, as the main dataset.

Attribute	Details
Dataset name	Air Quality Data in India (2015–2024)
Platform / publisher	Kaggle – ankushpanday1
File used	city_hour.csv (46.25 MB)
Size	438,245 rows and 16 columns
Cities	5 – Delhi, Mumbai, Chennai, Kolkata, Bangalore (87,649 rows each)
Time span	1 January 2015 to 31 December 2024, hourly records
AQI categories	6 – Good, Satisfactory, Moderate, Poor, Very Poor, Severe
Access method	kagglehub download in Google Colab

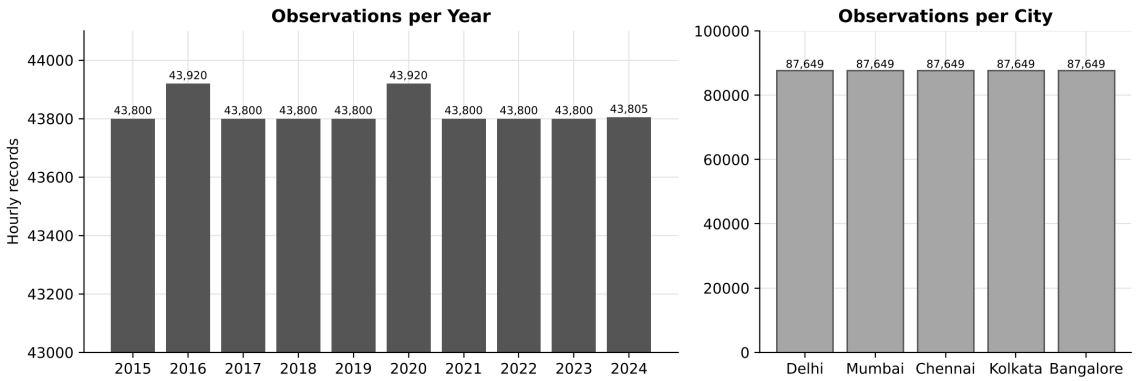


Figure 1. Number of hourly records per year and per city. Coverage is complete and identical for every city.

3.2 Automatic Column Discovery

The notebook does not assume fixed column names. It standardises them (for example `PM2.5` becomes `PM25`) and maps them to standard roles with regular expressions. All roles were found: `Datetime` → Date, `City`, `AQI`, `AQI_Bucket`, and all twelve pollutants (PM2_5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, Xylene).

3.3 Data Dictionary

Variable	Meaning	Type	Observed range
City	City of the observation	Categorical	5 cities
Date	Date and hour of observation	Datetime	2015–2024
PM2_5	Fine particulate matter $\leq 2.5\,\mu$m	Numerical	0 – 500
PM10	Particulate matter $\leq 10\,\mu$m	Numerical	0 – 600
NO / NO2 / NOx	Nitric oxide, nitrogen dioxide, nitrogen oxides	Numerical	0–200 / 0–150 / 0–250
NH3	Ammonia	Numerical	0 – 50
CO	Carbon monoxide	Numerical	0 – 10
SO2	Sulphur dioxide	Numerical	0 – 100
O3	Ozone	Numerical	0 – 200
Benzene / Toluene / Xylene	Volatile organic compounds	Numerical	0–20 / 0–30 / 0–10
AQI	Air Quality Index	Numerical	0 – 500
AQI_Bucket	AQI category	Categorical	6 categories

4. DATA PREPARATION AND DATA-QUALITY RESULTS

4.1 Data-Quality Checks

Check	Method	Result
Missing values	Count of nulls per column	0 in every column
Duplicate rows	`df.duplicated()`	0 duplicates
Invalid values	Negative or infinite pollutant/AQI values	0 found
Empty city names	Blank or null City	0 found
Unparseable dates	`pd.to_datetime(errors="coerce")`	0 rows dropped
Non-standard categories	Compare with the six CPCB labels	None
Outliers (IQR rule)	Below $Q_1 - 1.5\,$IQR or above $Q_3 + 1.5\,$IQR	0 in every column

Because the data contained no missing, duplicate or invalid values, the city-wise median imputation step left the shape unchanged at 438,245 $\times$ 16. For the models, imputation and scaling are still placed inside a scikit-learn `Pipeline` fitted on the training data only, so no test information can leak into training.

4.2 Outlier Analysis

Variable	Q1	Q3	IQR	Upper bound	Outliers
PM2.5	124.40	374.30	249.90	749.15	0
PM10	149.00	450.20	301.20	902.00	0
NO2	37.70	112.50	74.80	224.70	0
CO	2.50	7.50	5.00	15.00	0
AQI	125.40	374.90	249.50	749.15	0

No variable has a single outlier. Every column has hard limits (for example PM2.5 is exactly 0 to 500), and its quartiles sit almost exactly at one-quarter and three-quarters of that range. This is unusual for pollution data, which normally has a long right tail of extreme days, and is examined further in Section 7.

4.3 Feature Engineering

Feature	Source field	Meaning
Year, Month, Day	Date	Calendar position of the observation
Day_of_Week, Quarter	Date	Weekday (0 = Monday) and quarter (1–4)
Season	Month	Winter (Dec–Feb), Summer (Mar–May), Monsoon (Jun–Sep), Post-Monsoon (Oct–Nov)

The final model inputs are the 12 pollutants and 4 numeric calendar features (median-imputed and standardised), plus City and Season (one-hot encoded): 18 input columns in total. Leakage rules were applied throughout: the AQI category is never used to predict AQI, raw AQI is never used to predict the category, and the raw Date is never given to a model.

5. EXPLORATORY DATA ANALYSIS AND VISUALIZATIONS

Twelve visualizations were produced in the notebook with pandas, seaborn and matplotlib. Additional summary charts were generated from the notebook's printed outputs.

5.1 Summary Statistics

Variable	Mean	Median	Std	Min	Max
AQI	250.10	250.00	144.13	0	500
PM2.5	249.42	249.10	144.36	0	500
PM10	299.64	299.40	173.40	0	600
NO2	75.07	75.10	43.26	0	150
SO2	50.09	50.10	28.89	0	100
CO	5.00	5.00	2.89	0	10
O3	99.98	99.90	57.76	0	200

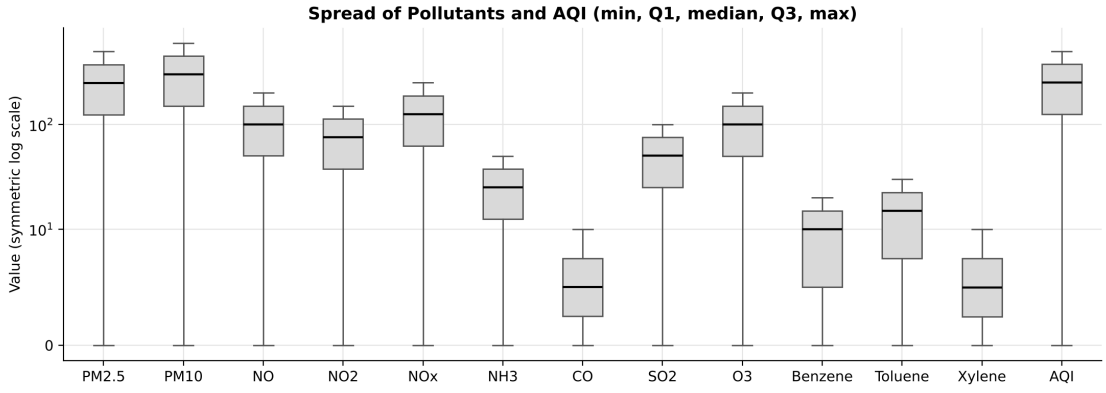
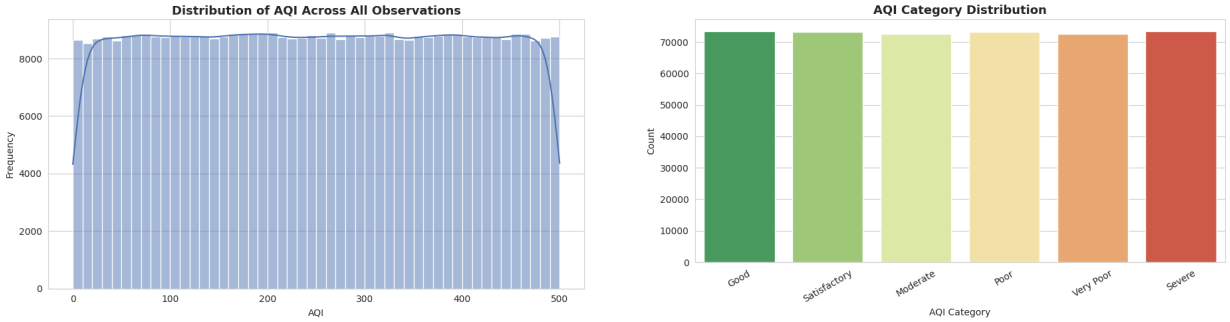


Figure 2. Spread of every pollutant and AQI (generated from the notebook's summary statistics).

Interpretation. For every variable the mean equals the median and both sit exactly at the middle of the range. AQI percentiles also rise in equal steps (5% = 25.4, 25% = 125.4, 50% = 250.0, 75% = 374.9, 95% = 474.8). This is the signature of a uniform distribution.

5.2 Visualizations 1 and 2 – AQI and AQI Category Distribution



(a) AQI distribution

(b) AQI category counts

Figure 3. Distribution of the regression and classification targets.

Interpretation. AQI is flat between 0 and 500 rather than right-skewed. The six categories are almost exactly equal: Good 73,450; Severe 73,324; Poor 73,172; Satisfactory 73,114; Very Poor 72,619; Moderate 72,566. The imbalance ratio is only 1.01, so class imbalance is not an issue here. In real Indian data, Severe hours are far rarer than Satisfactory or Moderate hours.

5.3 Visualization 3 – City-wise Average AQI

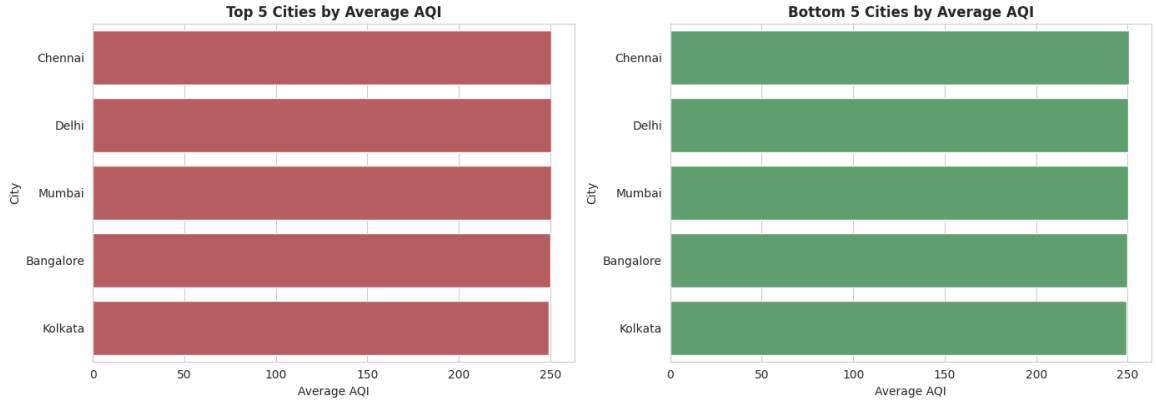


Figure 4. Cities ranked by average AQI. With only five cities, the top and bottom lists contain the same cities.

Interpretation. All five cities have an average AQI of about 250, so the ranking (Chennai highest, Kolkata lowest) is separated by only fractions of a point. This contrasts with real CPCB records, where Delhi's average AQI is typically far higher than Bangalore's or Chennai's.

5.4 Visualizations 4, 5 and 6 – AQI over Time, by Month and by Season

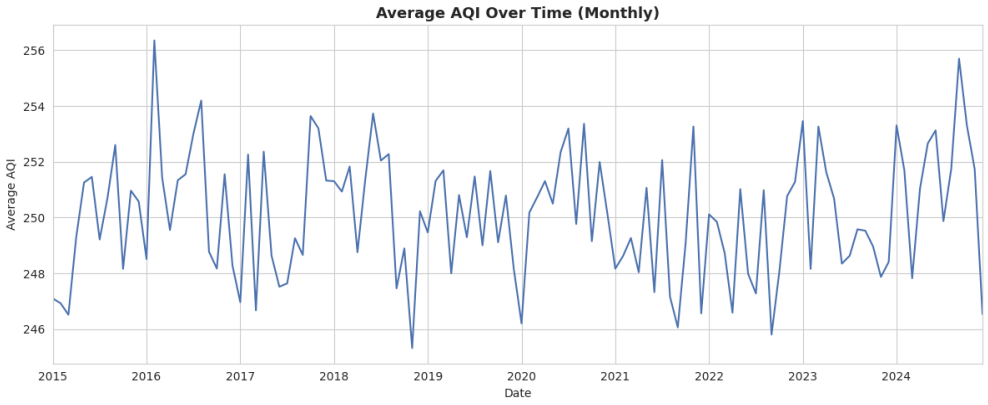
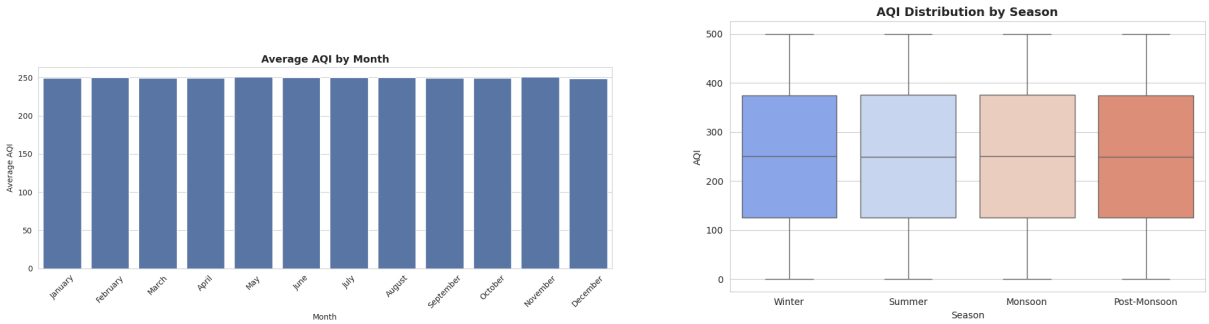


Figure 5. Average AQI over time (monthly average).



(a) Average AQI by month

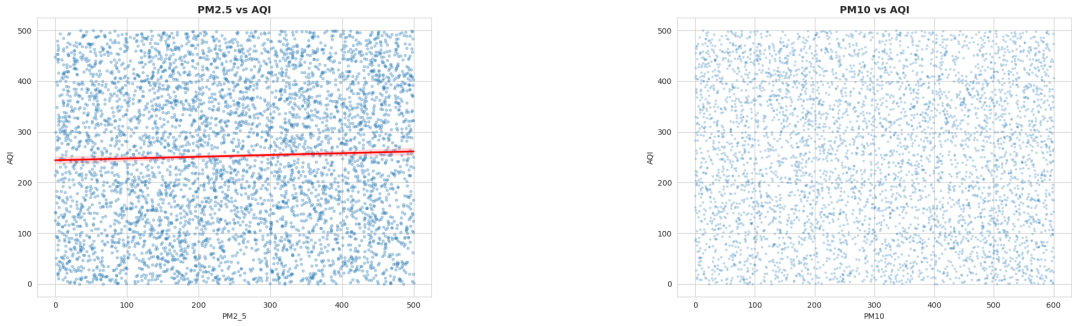
(b) AQI by season

Figure 6. Monthly and seasonal variation in AQI.

Season	Mean AQI	Median AQI	Records
Monsoon (Jun–Sep)	250.29	250.3	146,400
Post-Monsoon (Oct–Nov)	250.19	249.7	73,200
Summer (Mar–May)	250.13	249.7	110,400
Winter (Dec–Feb)	249.73	250.2	108,245

Interpretation. The monthly trend only fluctuates between about 245 and 256, with no repeating yearly cycle. Monthly averages range from 249.16 (December) to 250.93 (May), and the four seasons differ by just 0.56 AQI points. Monsoon is marginally the highest and Winter the lowest, which is the opposite of the real pattern in northern India, where winter inversions push AQI up and monsoon rain brings it down.

5.5 Visualizations 7 and 8 – PM2.5 and PM10 versus AQI



(a) PM2.5 vs AQI (b) PM10 vs AQI

Figure 7. Relationship between particulate matter and AQI (random sample of 5,000 points).

Interpretation. The points fill the whole square evenly and the fitted line is almost flat. A PM2.5 reading of 450 is as likely to come with AQI 20 as with AQI 480. In real data these plots show a tight rising band, because AQI is calculated from these pollutants.

5.6 Visualization 9 – Correlation Heatmap

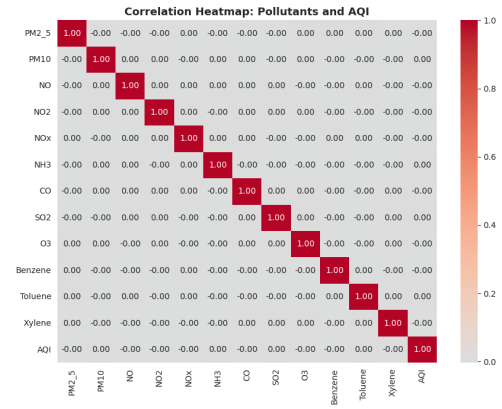
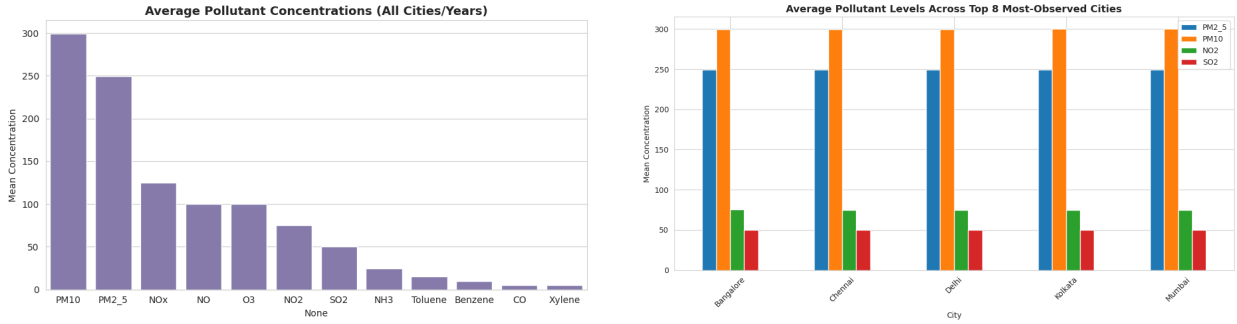


Figure 8. Correlation between pollutants and AQI.

Interpretation. Every off-diagonal correlation rounds to 0.00. AQI is uncorrelated with all twelve pollutants, and the pollutants are uncorrelated with each other. Even NO, NO_2 and NOx, which are chemically linked and share traffic sources in reality, show no relationship.

5.7 Visualizations 10 and 11 – Pollutant Levels

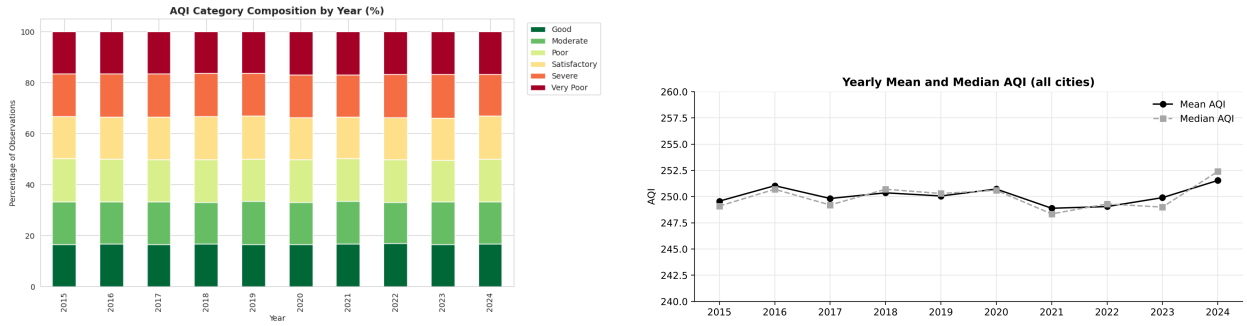


(a) Mean pollutant concentrations (b) Key pollutants by city

Figure 9. Average pollutant levels overall and by city.

Interpretation. PM10 (299.6) and PM2.5 (249.4) have the highest averages, each equal to half of its upper limit. The city chart shows identical pollutant profiles in all five cities.

5.8 Visualization 12 and Yearly Trend – Changes over the Years



(a) AQI category share by year

(b) Yearly mean and median AQI

Figure 10. Changes in air quality over the years 2015–2024.

Interpretation. Each category takes about one-sixth (16.7%) of observations in every year. Yearly mean AQI stays between 248.89 (2021) and 251.54 (2024), a range of under three points. There is no improvement, deterioration, or visible 2020 lockdown dip.

6. PREDICTIVE MODELLING AND EVALUATION

6.1 Experimental Setup

The data was split randomly into 80% training (350,596 records) and 20% testing (87,649 records) with random_state = 42. Each model was wrapped in a pipeline: median imputation and standard scaling for the 16 numeric features, and most-frequent imputation and one-hot encoding for City and Season.

Algorithm	Description	Status
Linear Regression	Linear baseline; fast and interpretable	Executed
Decision Tree	Splits data on feature thresholds; can model non-linear effects	Executed
Random Forest	Average of 200 trees on bootstrap samples	Not completed
Gradient Boosting	Trees built one after another to correct errors	Not completed

6.2 Evaluation Metrics

Metric	Formula	Interpretation
MAE	$\frac{1}{n}\sum \lvert y_i - \hat{y}_i \rvert$	Average error in AQI points
RMSE	$\sqrt{\frac{1}{n}\sum (y_i - \hat{y}_i)^2}$	Penalises large errors more heavily
R^2	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	1 = perfect; 0 = no better than the mean; below 0 = worse than the mean

6.3 AQI Regression Results (RQ1)

Model	MAE	RMSE	R^2
Linear Regression	124.770	144.059	−0.0003
Decision Tree	167.959	205.721	−1.0399

Table 2. Regression results on the 87,649-record test set.

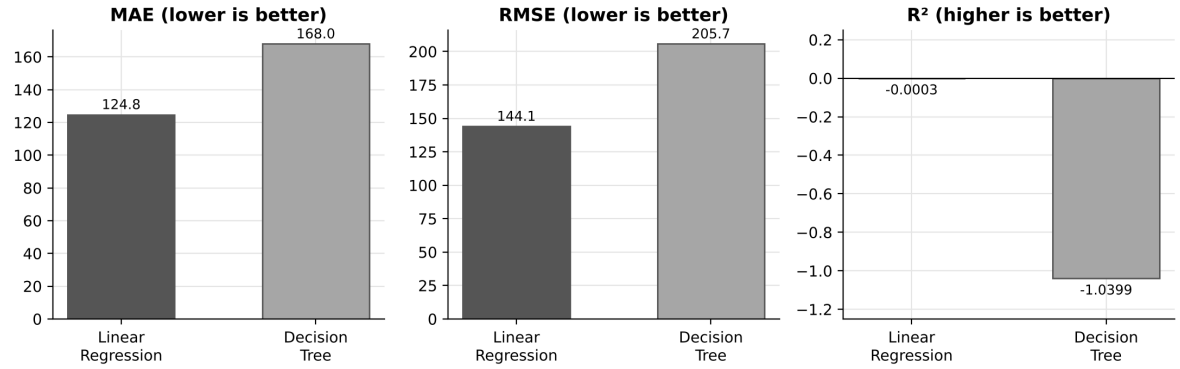


Figure 11. Regression performance of the two executed models (generated from the notebook output).

6.4 Interpretation of the Regression Results

Neither model learned anything useful, and both results match exactly what statistics predicts when AQI is unrelated to the inputs:

- **Linear Regression equals guessing the average.** For values spread uniformly between 0 and 500, the standard deviation is $500/\sqrt{12} = 144.3$ and the mean absolute distance from the centre is $500/4 = 125$. A model that always predicts 250 would therefore score RMSE $\approx$ 144 and MAE $\approx$ 125. Linear Regression scored 144.06 and 124.77, and $R^2 = 0$.
- **The Decision Tree memorised noise.** A fully grown tree fits each training point exactly. When the target is random, its prediction for a new point is effectively another random value, so the error variance doubles: RMSE $\approx \sqrt{2} \times 144 = 204$ and $R^2 \approx -1$. The tree scored 205.7 and -1.04.

The AQI-leakage concern raised in the notebook (that pollutants could "give away" AQI through the CPCB formula) does not apply to this dataset, because the formula relationship is absent.

6.5 Experiments that Did Not Complete

The notebook output ends after the Decision Tree. The Random Forest (200 trees on 350,596 rows) did not finish, most likely because of the Google Colab time or memory limit, so no later cell produced output. The table lists those experiments and the outcome expected from the evidence above. These expectations are reasoning, not measured results.

Experiment	Research question	Expected outcome given independent data
Random Forest, Gradient Boosting regression	RQ1	R^2 close to 0; averaging trees removes the Decision Tree's overfitting but cannot create signal
Six-class category classification	RQ2	Accuracy close to 1/6 = 16.7% (chance level)
Feature sets and City	RQ4, RQ5	No meaningful change in R^2
Feature importance	RQ3	Importance spread roughly evenly; permutation importance near 0
Early-warning classifier	RQ9	ROC-AUC close to 0.5
Lag-based forecasting	RQ10	R^2 close to 0 if hourly values are independent over time

7. IS THE DATASET REALISTIC? VALIDATION AGAINST THE CPCB FORMULA

Because the models failed in a way that exactly matches random data, the dataset itself was checked against what real air-quality data must look like.

Property	Expected in real CPCB data	Observed in city_hour.csv
AQI vs pollutants	AQI rises with the dominant pollutant (strong positive correlation)	Correlation 0.00 with every pollutant
Category vs AQI	Category follows the AQI breakpoints	Category does not match AQI (examples below)
Distribution shape	Right-skewed with rare extreme days	Flat (uniform) from 0 to the upper limit
Related pollutants	NO, NO_2 and NOx strongly correlated	Correlation 0.00
City differences	Delhi far more polluted than southern cities	All cities average ≈ 250
Seasonality	Winter peak, monsoon low in north India	Seasons differ by 0.56 points
Missing data	Sensor gaps and missing pollutants are common	No missing values at all

Category versus AQI. Among the ten rows printed by `df.head()` and `df.tail()`, none has a category that matches its AQI value under the CPCB breakpoints:

City	Date	AQI	Label in data	CPCB category
Delhi	01-01-2015	411.5	Moderate	Severe
Mumbai	01-01-2015	134.8	Poor	Moderate
Kolkata	01-01-2015	351.9	Good	Very Poor
Bangalore	01-01-2015	23.7	Moderate	Good
Chennai	31-12-2024	22.8	Severe	Good
Bangalore	31-12-2024	227.9	Good	Poor

Conclusion of the validation. Every column behaves like an independent random number drawn uniformly between fixed limits, and the AQI category appears to be assigned independently of AQI. The dataset is very likely synthetic. It is useful for practising a data pipeline, but it cannot answer questions about real air pollution in Indian cities. This is the most important finding of the project.

8. ANSWERS TO THE RESEARCH QUESTIONS

RQ	Answer from the outputs
RQ1	No, not on this dataset. Linear Regression $R^2 = -0.0003$ and Decision Tree $R^2 = -1.04$; AQI is unrelated to the inputs.
RQ2	Not executed. With categories independent of AQI and pollutants, accuracy near chance (16.7%) is expected.
RQ3	No pollutant predicts AQI: all twelve correlations are 0.00.
RQ4	Not executed; with zero correlation, adding pollutants cannot improve prediction.
RQ5	Not executed; all cities have the same AQI distribution, so City carries no information.
RQ6	All five cities average about 250; differences are fractions of a point.
RQ7	No meaningful variation: months range 249.16–250.93 and seasons 249.73–250.29.
RQ8	Stable: yearly mean AQI 248.89–251.54 between 2015 and 2024.
RQ9	Not executed; an early-warning model is expected to perform at chance level on this data.
RQ10	Not executed.

9. LIMITATIONS AND FUTURE SCOPE

9.1 Limitations

- The dataset appears to be synthetic: values are uniformly random and AQI and its category do not follow the CPCB formula, so findings do not describe real pollution.
- The notebook stopped after the Decision Tree; Random Forest, Gradient Boosting, classification, feature importance, early-warning and forecasting results were not produced.
- Only five cities are present, so city-level comparison is very limited.
- Weather, traffic and industrial data, which strongly affect real pollution, were not available.
- A random train-test split was used; for time-ordered data, a chronological split is more realistic.

9.2 Future Enhancements

- Repeat the analysis on official CPCB data, or on the widely used "Air Quality Data in India (2015–2020)" Kaggle dataset compiled from CPCB records.
- Add a validation step that recomputes AQI from pollutants and checks it against the given AQI before modelling.
- Train ensemble models on a sample (for example 50,000 rows) so they complete within Colab limits.
- Add meteorological data and use time-series cross-validation and sequence models for forecasting.
- Build an early-warning dashboard once a realistic dataset is in place.

10. CONCLUSION

This project applied a complete predictive-analytics workflow to the Kaggle dataset "Air Quality Data in India (2015–2024)": automatic column discovery, data-quality checks, outlier analysis, feature engineering, twelve exploratory visualizations, and leakage-safe regression pipelines.

The dataset was perfectly clean, with no missing values, duplicates or outliers, but that cleanliness turned out to be a warning sign. Every pollutant and AQI value is spread evenly between fixed limits, all correlations are zero, and cities, months, seasons and years are indistinguishable. Linear Regression performed exactly like predicting the average ($R^2 = -0.0003$), and the Decision Tree performed exactly like a model fitting random noise ($R^2 = -1.04$). The AQI categories also do not match the AQI values.

The main lesson is that data must be validated before it is modelled. A model's failure can reveal a problem with the data rather than with the algorithm, and checking results against domain knowledge (here, the CPCB AQI formula) is what exposed it. The workflow built in this project is ready to be reused on a genuine CPCB dataset, where pollutants do drive AQI and an early-warning system for Indian cities becomes realistic.

11. TOOLS, REPRODUCIBILITY AND REFERENCES

11.1 Tools and Technologies

Tool / Library	Purpose in the project
Google Colab	Cloud notebook environment for running the analysis
Python 3	Programming language for data processing and modelling
kagglehub	Downloads the dataset from Kaggle
pandas, NumPy	Loading, cleaning, transforming and aggregating data
matplotlib, seaborn	Charts and heatmaps
scikit-learn	Pipelines, encoding, train/test split, models and metrics

11.2 Steps to Reproduce

- Open Google Colab and upload Early_Prediction_of_Poor_Air_Quality_in_Indian_Cities.ipynb.

- Run the cells in order; sign in to Kaggle if kagglehub asks for authentication.
- The dataset is downloaded with:

```
import kagglehub
path = kagglehub.dataset_download("ankushpanday1/air-quality-data-in-india-2015-2024")
```

11.3 References

- Central Pollution Control Board (CPCB). National Air Quality Index. Ministry of Environment, Forest and Climate Change, Government of India, 2014.
- Kaggle. Air Quality Data in India (2015–2024), published by ankushpanday1. https://www.kaggle.com/datasets/ankushpanday1/air-quality-data-in-india-2015-2024
- Breiman, L. Random Forests. Machine Learning, 45(1), 5–32, 2001.
- Pedregosa, F. et al. Scikit-learn: Machine Learning in Python. JMLR, 12, 2825–2830, 2011.
- World Health Organization. WHO Global Air Quality Guidelines. Geneva, 2021.
- pandas, matplotlib and seaborn documentation: https://pandas.pydata.org, https://matplotlib.org, https://seaborn.pydata.org

ACKNOWLEDGEMENT

We sincerely thank Mrs. V. Vasuki Rohini Devi for her guidance and academic support throughout this project. We also acknowledge CBIT, the developers of the open-source Python libraries used in this study, and the Kaggle dataset publisher for making the data available for educational analysis.